GPE_EXT_001 — Gross–Pitaevskii GPU Numerical Benchmark

SST-Workbench, numerical development protocol

v0.1.1 / Framework v1.0.10

1 Scope

This is a numerical GPE benchmark, not an SST/PKLSA gravity, particle, clock or Euler-flow validation. Framework 1.0.10 is not yet canonically promoted. No physical inference from a source checksum is permitted.

2 Research question

Can periodic dimensionless cubic GPE split-step Fourier dynamics reproduce exact solutions, numerical invariants, second-order temporal convergence and Intel Arc XPU screening parity?

3 Hypotheses

H_0: pre-registered numerical errors or empirical splitting order are unacceptable. H_1: the stated numerical thresholds are satisfied. These are engineering hypotheses, not SST confirmations.

4 Assumptions

Periodic square computational domain; Fourier spectral discretization; dimensionless real cubic coupling $g > 0$; conserved particle norm; no damping or trapping potential; XPU screening complex64 versus independent NumPy complex128. Physical condensate healing length and boundary physics are outside these tests.

5 Symbols and dimensions

Let $\mathbf{x} = \mathbf{x}_{\mathrm{SI}}/L_0$, $t = t_{\mathrm{SI}}/T_0$, $\psi = \Psi/\Psi_0$; L_0, T_0, Ψ_0 are declared normalization scales, so the implemented model is fully dimensionless. No SST constants are fitted here.

6 Equations and numerical discretization

$$i\partial_t\psi = -\frac{1}{2}\nabla^2\psi + g(|\psi|^2 - n_0)\psi. \tag{1}$$

$$\psi_{\mathrm{PW}} = A\exp\left[i(\mathbf{k}\cdot\mathbf{x} - \omega t)\right], \quad \omega = \frac{|\mathbf{k}|^2}{2} + g(A^2 - n_0). \tag{2}$$

$$\psi^{n+1} = \exp[-i\tfrac{\Delta t}{2}g(|\psi^*|^2 - n_0)]\,\mathcal{F}^{-1}[e^{-i\Delta t|\mathbf{k}|^2/2}\mathcal{F}\{e^{-i\Delta t g(|\psi^n|^2-n_0)/2}\psi^n\}], \tag{3}$$

where ψ^* denotes the state after the kinetic substep; $\mathcal{F}$ is the discrete periodic FFT. Same units appear in each exponential (dimensionless phase).

7 Algorithm

Source download and checksum; NumPy FP64 analytic plane wave; norm and reverse propagation; fixed final-time dt-refinement; Intel XPU complex64 split-step full FFT parity; framework ledger creation and immutability checks. A numerical fine-step solution is used for order estimation and is not a measured physical ground truth.

8 External source

J. Kwon and Y.-i. Shin, *Data and analysis scripts for "Dynamic similarity of vortex shedding in a superfluid flowing past a penetrable obstacle"*, Zenodo, 2026, `https://doi.org/10.5281/zenodo.20068724`. The 18.7 MB file contains *processed observables*, not complete time-dependent ψ snapshots. Archive MD5 and ZIP CRC are checked; source-field parity is untested.

9 Preregistered gates

G_0: framework frozen bundle. G_1: checksum plus CRC. G_2: analytic relative L^2 error $< 10^{-10}$, norm drift $< 10^{-10}$, reverse-step error $< 10^{-10}$. G_3: temporal convergence observed minimum order > 1.75 with fixed final time and 128-step surrogate. G_6: genuine Torch XPU/NumPy relative $L^2 < 7 \cdot 10^{-4}$ and GPU norm drift $< 7 \cdot 10^{-4}$. G_4, G_5, G_7 intentionally disabled; G_8 unresolved, G_9 deferred until separate reveal. G_6 is screening only, not SYCL or native whole-solver parity.

10 Numerical uncertainties and limits

Floating-point effects, FFT conventions, periodic aliasing and GPU driver support. $\Delta t \to 0$ convergence is estimated at fixed spatial resolution; temporal and spatial convergence must be separated for later physical tests. Phase winding, reconnection and actual PKLSA data are not currently checked.

11 Backend authority

NumPy complex128 is numerical reference; Torch XPU complex64 is screening only. A missing GPU is an unavailable prerequisite, never an implicit CPU parity PASS. DD32/native SYCL whole-solver parity is not claimed.

12 Blindness

Pre-registered numerical gates and thresholds are frozen ahead of validation. No secret labels or target-varying choices are present in this stage. No automatic REVEAL; private commitments retained in framework. The external dataset is not a secret challenge set.

13 Statistics

Deterministic convergence and tolerance tests, not a hypothesis test with random p-values. Independent sampling unit for eventual physical replication must be a genuinely independent simulation/experiment family, not additional timesteps from the same simulation.

14 Automatically generated evidence

Machine-readable science contract

Research question Can a periodic dimensionless Gross-Pitaevskii equation split-step Fourier solver reproduce exact and convergent reference dynamics on independent NumPy FP64 and Intel Arc XPU FP32 implementations?

Objective Certify numerical checks and XPU screening only; archive downloaded external processed-observable data with provenance. No SST/Euler/PKLSA physical equivalence is asserted.

H_0 At least one preregistered numerical reference or convergence criterion fails; GPE solver numerical qualification is denied.

H_1 All completed CPU reference checks and the temporal convergence check satisfy their independently specified thresholds; optional Intel XPU screening also satisfies parity thresholds.

Registered equations

E1: Continuous dimensionless conservative GPE

$$i\,\partial_t\psi = -\frac{1}{2}\nabla^2\psi + g(|\psi|^2 - n_0)\psi \tag{4}$$

Dimensional check: All variables dimensionless after xSI/L0 and tSI/T0 normalization

E2: Exact periodic plane wave reference

$$\psi = Ae^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}, \quad \omega = |\mathbf{k}|^2/2 + g(A^2 - n_0) \tag{5}$$

Dimensional check: omega and all exponents dimensionless

E3: Relative complex L2 error

$$\epsilon_2 = \|\psi_{\text{candidate}} - \psi_{\text{reference}}\|_2 / \|\psi_{\text{reference}}\|_2 \tag{6}$$

Dimensional check: Ratio of equal-dimension norms is dimensionless

E4: Measured temporal order

$$p = \log_2(\epsilon_{\Delta t}/\epsilon_{\Delta t/2}) \tag{7}$$

Dimensional check: Logarithm of dimensionless error ratio

Registered code-to-mathematics steps

1. **S1** [G1; equations E1]: Retrieve Zenodo public source and verify published MD5 and all ZIP member CRC
2. **S2** [G2; equations E1, E2, E3]: Propagate exact plane wave and reverse a nonuniform complex field using NumPy FP64
3. **S3** [G3; equations E1, E3, E4]: Split-step dt refinement at fixed final time with independent fine-step surrogate
4. **S4** [G6; equations E1, E3]: Propagate same initial field using Intel XPU complex64 and compare to independent NumPy complex128

Automatic run provenance

Field	Recorded value
Run ID	20261019T233552643973Z-f6254125
Run phase	BLIND
Protocol bundle SHA-256 (byte exact)	cfb2c9be243cc5c4cadb7001c3342867c2307d42957707b4 4bb51562360a0dc4
Portable protocol diagnostic SHA-256	a9f2a99f29a2a04fdf11577702b90f7e6dcbb812acf9113c 82b412ce37aaa9aa
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Science evidence SHA-256	d81fe015481ea417bd83a0a06df4153d88175b0463758f7e bf0b7b249749c78e
Final output manifest	See OUTPUT_MANIFEST.json (finalized after PDF and report metadata)

Gate ledger

Gate	Status	Question	Reason
G0	PASS	Frozen protocol and v1.0.10 runtime verified?	SST framework verified the frozen protocol before invoking pipeline.
G1	PASS	Is the external Zenodo processed-observable archive checksum and CRC verified?	Published Zenodo MD5 + ZIP CRC; not a wavefunction reference
G2	PASS	Does an independent NumPy-FP64 GPE TSSP pass plane-wave, norm and time-reversal checks?	Periodic plane-wave exact solution, mass invariance, time reversal
G3	PASS	Is second-order time-step convergence demonstrated without using physical field labels?	Strang splitting dt refinement at fixed final time; numerical surrogate, not physical discovery
G4	NOT_APPLICABLE	Is there a separate field-level independent research dataset confirmation? (disabled pending comparable fields)	Disabled by framework profile
G5	NOT_APPLICABLE	Does an independent native C++ FP64 GPE solver certify parity? (disabled: unavailable)	Disabled by framework profile
G6	PASS	Does the Intel XPU real GPU solver agree with NumPy FP64 under preregistered screening thresholds?	Actual Intel GPU FFT/ pointwise GPE solver compared to independent NumPy FP64
G7	NOT_APPLICABLE	Do multiple independent sources replicate? (disabled pending data mapping)	Disabled by framework profile
G8	UNRESOLVED	Does genuine PKLSA/Euler vortex mechanism reproduce GPE measurements? (unimplemented)	No scientific evaluator implemented for this gate; not executed

G9	DEFERRED	Separate blind reveal integrity verification (not an automatic physical claim)	Reveal is a separate, commitment-verified phase and is not executed during blind runs.

Scientific qualification (separate from measured gate outcomes)

Gate	Evidence qualification	Blocking prerequisites
G0	QUALIFIED	none
G1	QUALIFIED	none
G2	QUALIFIED	none
G3	QUALIFIED	none
G4	NOT_APPLICABLE	none
G5	NOT_APPLICABLE	none
G6	QUALIFIED	none
G7	NOT_APPLICABLE	none
G8	UNRESOLVED	none
G9	DEFERRED	G8

Backend manifest


```
{
  "dataset": {
    "bytes": 18720026,
    "checksum_ok": true,
    "crc_ok": true,
    "doi": "10.5281/zenodo.20068724",
    "file_count": 46,
    "sha256":
"69a58532fc01a013a2a7ba17428787f68e2c62b4a5181cee4e822092d7aa5b9e",
    "sha256_match": true,
    "verified": true
  },
  "gpe_scheme": "strang-split-step-FFT-periodic",
  "gpu": {
    "actual_backend": "pytorch-xpu-fp32",
    "device_name": "Intel(R) Arc(TM) A770 Graphics",
    "gpu_cpu_relative_l2": 3.956597249386977e-06,
    "gpu_norm_relative_drift": 7.271766662597656e-06,
    "n": 64,
    "reference_kind": "independent numpy FFT fp64 implementation",
    "requested_backend": "pytorch-xpu-fp32",
    "ring3d": {
      "gpu_cpu_relative_l2": 1.170856082278945e-06,
      "n": 40,
      "seed_kind":
"illustrative_closed_ring_not_relaxed_or_validated_topology",
      "steps": 8
    },
    "steps": 48,
    "torch_version": "2.14.1+xpu",
    "units": "dimensionless"
  },
  "no_pklsa_equivalence_claim": true,
  "python_reference": {
    "actual_backend": "numpy-fp64",
    "authority": "NUMERICAL_REFERENCE"
  },
  "schema": "SST-BACKEND-MANIFEST-2",
  "status": "DEVELOPMENT_NOT_PHYSICAL_CERTIFICATION"
}
```

Source manifest


```
{
  "manifest_sha256":
"720ef38019b90186730fa5d74e1b75370ba0b762b33056e6f5839e6bcb1dc6eb",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "PUBLIC-ZENODO-20068724",
      "evidence_class": "independent_source",
      "exists": true,
      "independence_family": "Kwon-Shin-vortex-shedding",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\F_explora
tory\\GPE_EXT_001_GPE_GPU_Benchmark_v0.1.1\\data\\external\\Vortex-shedding-P
RR-data-1.0.0.zip",
      "provenance_family": "Zenodo-20068724",
      "registry_hint": {
        "catalog_id": "PUBLIC-ZENODO-20068724",
        "hits": []
      },
      "role": "external processed-observable source",
      "sha256":
"69a58532fc01a013a2a7ba17428787f68e2c62b4a5181cee4e822092d7aa5b9e",
      "source_id": "ZENODO-20068724",
      "version": "v1.0.0"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```


15 Permitted interpretation

Numerical PASS supports implementation qualification only. Source PASS supports data integrity only. Neither demonstrates the Euler–PKLSA limit nor the emergence of SST gravity/time dilation. Failure of GPE numerical checks is not a falsification of SST.

16 Reproducibility

Run `run_all.cmd BASIC` or `run_all.cmd FULL`; use the exact framework v1.0.10 directory. Check GATE_LEDGER.json, BACKEND_MANIFEST.json, SOURCE_MANIFEST.json, independent checksum ledger, PDF and BLIND ZIP. GPE physical units cannot be recovered without specifying L_0, T_0, Ψ_0 and a material coupling.

References

[1] E. P. Gross (1961), "Structure of a Quantized Vortex in Boson Systems," *Il Nuovo Cimento* **20**, 454–477. doi:10.1007/BF02731494.

[2] L. P. Pitaevskii (1961), "Vortex Lines in an Imperfect Bose Gas," *Soviet Physics JETP* **13**, 451–454. `http://www.jetp.ras.ru/cgi-bin/dn/e_013_02_0451.pdf`.

[3] W. Bao, D. Jaksch, and P. A. Markowich (2003), "Numerical Solution of the Gross–Pitaevskii Equation for Bose–Einstein Condensation," *Journal of Computational Physics* **187**, 318–342. doi:10.1016/S0021-9991(03)00102-5.

[4] J. Kwon and Y.-i. Shin (2026), "Data and analysis scripts for Dynamic similarity of vortex shedding in a superfluid flowing past a penetrable obstacle," *Zenodo*. doi:10.5281/zenodo.20068724.

[5] ETH Zurich Quantum Optics Group, "TorchGPE," research software. https://github.com/qo-eth/TorchGPE.